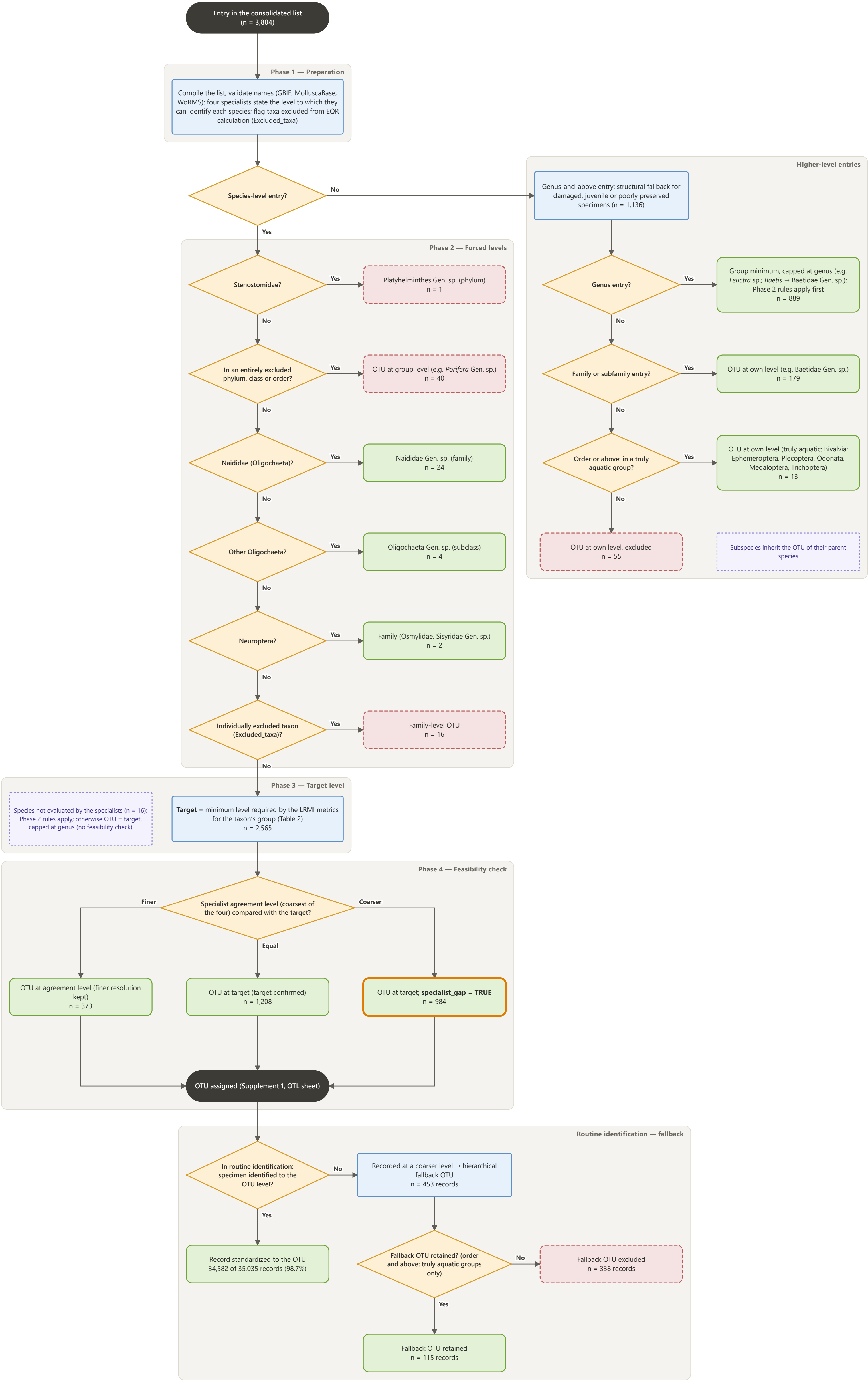


Decision tree for assigning operational taxonomic units (OTUs)

Branch counts are species evaluated by the four specialists (n = 2,652), higher-level entries of the operational taxalist, or LEPA biomonitoring records 2014–2022, as labelled.

Start / result Action Decision (yes / no) OTU retained for EQR calculation OTU excluded from EQR calculation OTU finer than specialist agreement (specialist_gap = TRUE) Annotation



Worked examples

Taxon	Group	Specialists 1–4	Path through the tree	OTU
<i>Dicranota bimaculata</i>	Diptera	s · o · g · g	Species → no Phase 2 rule → target family (Diptera) → agreement order is coarser → OTU at target, specialist_gap = TRUE	Pediciidae Gen. sp.
<i>Paraleptophlebia submarginata</i>	Ephemeroptera	s · s · s · g	Species → no Phase 2 rule → target family (Ephemeroptera) → agreement genus is finer → OTU at genus	Paraleptophlebia sp.
<i>Leptophlebia marginata</i>	Ephemeroptera	s · s · f · g	Species → no Phase 2 rule → target family (Ephemeroptera) → agreement family equals target → OTU at family	Leptophlebiidae Gen. sp.
<i>Stylaria lacustris</i>	Oligochaeta	s · sc · s · sc	Species → Phase 2: Naididae → OTU forced to family (skips Phases 3–4)	Naididae Gen. sp.
Specimen identified only as “Diptera”	Diptera	—	Routine identification: not identified to family → fallback Diptera Gen. sp. (order) → Diptera not truly aquatic → excluded from EQR	Diptera Gen. sp.

Specialist levels: s species, g genus, f family, o order, sc subclass. The agreement level is the coarsest level reported by any specialist. Each example is verified against the OTU assigned by script 2.

Reference table — minimum level required by the LRMI metrics, by group

Group	DSFI requirement	BMWP/ASPT (all scored taxa)	Other metrics	Resulting minimum
Malacostraca	<i>Asellus</i> , <i>Gammarus</i> at genus (IG + diversity)	Score 8: Astacidae [f]. Score 6: Gammaridae [f], Corophiidae [f], Crangonyctidae [f], Niphargidae [f], Pontogammaridae [f]. Score 3: Asellidae [f]. Excluded taxa (Talitridae, non-Asellidae Isopoda, <i>Neomysis/Praunus</i> , Crangonidae/Palaemonidae) → family level, excl. EQR.	%CrHi pools Malacostraca	Genus for <i>Asellus</i> , <i>Gammarus</i> ; family for others
Gastropoda	<i>Ancylus</i> , <i>Lymnaea</i> at genus (diversity)	Score 6: Viviparidae [f], Neritidae [f], Ancyliidae [f], <i>Ancylus</i> [g], Acroloxidae [f]. Score 5: Hydrobiidae [f]. Score 3: Lymnaeidae [f], Physidae [f], Planorbidae [f], Valvatidae [f], Bithyniidae [f].	—	Genus for <i>Ancylus</i> , <i>Lymnaea</i> ; family for others
Bivalvia	<i>Sphaerium</i> at genus (diversity)	Score 6: Unionidae [f]. Score 3: Sphaeriidae [f].	—	Genus for <i>Sphaerium</i> ; family for others
Plecoptera	13 genera at genus (IG1/IG2). Families: Leuctridae, Capniidae, Perlodidae, Perlidae, Chloroperlidae (IG1); Taeniopterygidae, Nemouridae (IG2).	Score 10: Taeniopterygidae [f], Leuctridae [f], Perlodidae [f], Perlidae [f], Chloroperlidae [f], Capniidae [f]. Score 7: Nemouridae [f].	#DEP: species	At least genus
Ephemeroptera	8 families: Ephemeridae (IG1), Ametropodidae, Ephemerellidae, Heptageniidae, Leptophlebiidae, Siphonuridae (IG2), Caenidae (IG3), Baetidae (IG5).	Score 10: Siphonuridae [f], Heptageniidae [f], Leptophlebiidae [f], Ephemerellidae [f], Potamanthidae [f], Ephemeridae [f]. Score 7: Caenidae [f]. Score 4: Baetidae [f].	#DEP: species	At least family
Trichoptera	<u>Case-bearing</u> : Beraeidae, Brachycentridae, Goeridae, Glossosomatidae, Hydroptilidae, Leptoceridae, Lepidostomatidae, Limnephilidae, Molannidae, Odontoceridae, Phryganeidae, Sericostomatidae. <u>Caseless</u> : Ecnomidae, Hydropsychidae, Philopotamidae, Polycentropodidae, Psychomyiidae, Rhyacophilidae.	Score 10: Beraeidae [f], Brachycentridae [f], Goeridae [f], Lepidostomatidae [f], Leptoceridae [f], Molannidae [f], Odontoceridae [f], Phryganeidae [f], Sericostomatidae [f]. Score 8: Ecnomidae [f], Philopotamidae [f], Psychomyiidae [f]. Score 7: Apataniidae [f], Glossosomatidae [f], Limnephilidae [f], Polycentropodidae [f], Rhyacophilidae [f]. Score 6: Hydroptilidae [f]. Score 5: Hydropsychidae [f].	—	At least family
Coleoptera	<i>Elmis</i> , <i>Limnius</i> , <i>Elodes</i> at genus (IG1/IG2 + diversity)	Score 5: Haliplidae [f], Hygrobiidae [f], Noteridae [f], Dytiscidae [f], Gyrinidae [f], Hydrophilidae [f], Hydraenidae [f], Hydrochidae [f], Helophoridae [f], Georissidae [f], Clambidae [f], Scirtidae [f], Helodidae [f], Dryopidae [f], Elmidae [f], Chrysomelidae [f], Curculionidae [f].	—	Genus for <i>Elmis</i> , <i>Limnius</i> , <i>Elodes</i> ; family for others
Diptera	<i>Chironomus</i> at genus ; Eristalinae at subfamily ; Simuliidae, Psychodidae, Chironomidae at family	Score 5: Tipulidae [f], Simuliidae [f], Limoniidae [f], Pediciidae [f], Cylindrotomidae [f]. Score 2: Chironomidae [f].	#DEP: family	Family ; genus for <i>Chironomus</i> ; subfamily for Eristalinae
Megaloptera	<i>Sialis</i> at genus (IG4 + neg. diversity)	Score 4: Sialidae [f].	—	Genus for <i>Sialis</i>
Hirudinea	<i>Erpobdella</i> , <i>Helobdella</i> at genus (neg. diversity)	Score 4: Piscicolidae [f]. Score 3: Glossiphoniidae [f], Hirudinidae [f], Erpobdellidae [f].	%CrHi pools Hirudinea	Genus for <i>Erpobdella</i> , <i>Helobdella</i> ; family for others
Oligochaeta	Naididae at family (IG6; formerly Tubificidae); Oligochaeta ≥ 100 individuals as precluder	Score 1: Oligochaeta [sc], Naididae [f].	—	Naididae at family ; others at subclass
Hemiptera	—	Score 10: Aphelocheiridae [f]. Score 5: Nepidae [f], Naucoridae [f], Notonectidae [f], Pleidae [f], Corixidae [f], Gerridae [f], Hydrometridae [f], Mesoveliidae [f].	%EHP pools Hemiptera. Excluded families → family-level OTU.	At least family
Odonata	—	Score 8: Lestidae [f], Calopterygidae [f], Gomphidae [f], Cordulegastridae [f], Aeshnidae [f], Corduliidae [f], Libellulidae [f]. Score 6: Coenagrionidae [f], Platynemididae [f].	—	At least family
Turbellaria	Tricladida at order (pos. diversity)	Score 5: Planariidae [f], Dendrocoelidae [f], Dugesiidae [f].	—	At least family
Lepidoptera	—	—	—	At least family ^a
Neuroptera	—	—	—	At least family (Phase 2)

[f] family, [g] genus, [sc] subclass. Minimum levels are floors, not ceilings: if all four specialists can identify a taxon to a finer level, the finer level is kept (Phase 4).

^a Lepidoptera are not used directly by any LRMI metric. A family-level minimum was nevertheless set because order-level Lepidoptera entries are excluded under the truly aquatic rule; an order-level OTU would therefore remove aquatic Crambidae and Pyralidae from the total sample abundance, which forms the denominator of %EHP~%CrHi.